

Reader Report: Week 06

Probabilistic Invariance: VI-BIP and the Core Theorems

Daniel Hardesty Lewis (dl3645)

2026 Spring

Summary

Wu and Blei (2025) place a prior over binary selectors $z \in \{0, 1\}^p$, where $z_j = 1$ indexes a feature in the invariant set S^* (Peters et al., 2016). The posterior $p(z \mid \mathcal{D}) \propto p(z) \prod_{e,i} \hat{g}(Y_i^e \mid X_{ei}^z) / \hat{p}_e(Y_i^e \mid X_{ei}^z)$ rewards selectors for which the pooled conditional $\hat{g}$ dominates each environment-specific conditional $\hat{p}_e$. Consistency (Thm 1) requires $\mu_{\min} > 0$, the minimum KL divergence between $\hat{g}$ and any $\hat{p}_e$ over non-invariant features (Wu and Blei, 2025, Def. 1). Contraction (Thm 2) gives $\text{TV}(p(z \mid \mathcal{D}), \delta_{z^*}) = O(2^p e^{-\kappa n E \mu_{\min}})$. Both bounds assume squared loss under a linear model, and the product likelihood weights all environments symmetrically. VI-BIP (Wu and Blei, 2025, Sec. 4) approximates the posterior via mean-field variational inference (Blei et al., 2017), optimizing the ELBO with the U2G gradient estimator.

Critique

Peters et al. (2016), Fan et al. (2024), and Wu and Blei (2025) all enforce the same invariance condition under squared loss, yet what separates them is what the practitioner assumes about the environments before seeing data. BIP’s symmetric product likelihood weights every environment equally, so its contraction rate $nE \gtrsim O(p/\mu_{\min})$ is bottlenecked by $\mu_{\min}$, the minimum KL divergence between pooled and local conditionals of $Y \mid X$. NegDRO measures heterogeneity differently: its identification requires $\lambda > 0$, a spectral condition on the second moments of X across environments (Fan et al., 2024, Cond. 1b). Its tuning parameter γ controls how negative the environment weights can go (Fan et al., 2024, Eq. 9), with $\gamma = 0$ recovering the symmetric simplex. A dataset can satisfy one heterogeneity condition but not the other, so the two frameworks are not merely different algorithms for the same problem.

Yet no theory characterizes where the boundary lies. There is no decision rule telling a practitioner when symmetric weighting wastes known structure, or when asymmetric weighting overfits to a misidentified dominant environment. The gap is acknowledged by neither paper.

The experiments do not close this gap. Wu and Blei (2025, Sec. 5) validate VI-BIP exclusively on CRISPR knockouts where $\mu_{\min} > 0$ holds by construction, the one regime where a practitioner *could* justify asymmetric weighting and BIP’s agnosticism is least needed. The empirical case for VI-BIP in observational or low-intervention settings, where symmetric weighting should matter most, remains unmade.

Questions

Both questions below concern whether the practitioner-knowledge gap between BIP and NegDRO can be bridged.

1. NegDRO’s rate improves smoothly in γ (Fan et al., 2024, Prop. 2), but this characterizes NegDRO alone. Is there a regime—characterized by the number of environments E , the ratio $\lambda/\mu_{\min}$, or the practitioner’s uncertainty about which environment dominates—in which BIP’s symmetric posterior provably outperforms NegDRO’s asymmetric point estimate, or vice versa?
2. Nonparametric extensions of ICP exist for the symmetric case (Heinze-Deml et al., 2018), but no nonparametric analogue of NegDRO’s asymmetric weighting has been proposed. Does the practitioner-knowledge distinction identified above survive when the loss is no longer quadratic, or does the symmetry-versus-asymmetry tradeoff collapse under nonparametric conditionals?

References

- D. M. Blei, A. Kucukelbir, and J. D. McAuliffe. Variational inference: A review for statisticians. *Journal of the American Statistical Association*, 112(518):859–877, 2017.
- Y. Fan, S. Li, and Z. Yang. Causal invariance learning via efficient nonconvex optimization. *arXiv preprint arXiv:2412.11850*, 2024. URL <https://arxiv.org/abs/2412.11850>.
- C. Heinze-Deml, J. Peters, and N. Meinshausen. Invariant causal prediction for nonlinear models. *Journal of Causal Inference*, 6(2), 2018. doi: 10.1515/jci-2017-0016.
- J. Peters, P. Bühlmann, and N. Meinshausen. Causal inference by using invariant prediction: identification and confidence intervals. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 78(5):947–1012, 2016.
- Y. Wu and D. M. Blei. Bayesian invariance modeling of multi-environment data. *arXiv preprint arXiv:2506.22675*, 2025. URL <https://arxiv.org/abs/2506.22675>.